SEBASTIÁN AYALA-RUANO

I am a researcher and freelance data scientist with six years of experience in **bioinformatics** and **cheminformatics** at various laboratories in Ecuador, South Korea, the Netherlands, and Denmark. My current research interests are focused on **network science**, **deep learning**, and **software development** to study **complex biological systems**. I am an active member of several **research, open-science, and software development communities** (The Turing Way, Open Life Science, Streamlit Creators, and ISCBSC) and have led initiatives to advance bioinformatics in Ecuador and Latin America.

EDUCATION

2024 | 2022

MSc, Systems Biology

Maastricht University (UM) — Maastricht, the Netherlands

- **Scholarship**: UM Holland-High Potential Scholarship (non-EU/EEA students).
- **GPA**: 8.78/10 (**Cum Laude**) - highest score in the Systems Biology 2024 class.
- **Thesis** (MaCSBio thesis award): MicW2Graph - Building a knowledge graph of the wastewater treatment microbiome and its biological context.

2020 | 2016

B.Eng., Biotechnology

Universidad San Francisco de Quito (USFQ) — Quito, Ecuador

- **Minor**: Software engineering.
- **GPA**: 3.78/4 (**Magna Cum Laude**) - second highest score in the College of Biological and Environmental Sciences 2020 class.

WORK EXPERIENCE

Sep. 2025 | Oct. 2024

Research Assistant

Multi-omics Network Analytics Group - Novo Nordisk Foundation Center for Biosustainability, Technical University of Denmark — Copenhagen, Denmark

- Contributed to the development of **bioinformatics software tools**, including VueGen, VueCore, ABaCo, and ACore, a suite of **Python packages** that automate report generation, visualization, batch correction, and analysis of multi-omics data.
- **Advisor**: Alberto Santos Delgado

May. 2024 | Nov. 2023

Research Intern

Multi-omics Network Analytics Group - Novo Nordisk Foundation Center for Biosustainability, Technical University of Denmark — Copenhagen, Denmark

- Investigated the wastewater treatment microbiome and developed **MicW2Graph**, an **open-source knowledge graph** integrating metagenomic and metatranscriptomic data with their biological context, including biological processes, environmental features, chemical compounds, and additional metadata (More details).
- **Advisors**: Alberto Santos Delgado, Martina Kutmon

Oct. 2022 | Apr. 2022

Data Science Consultant

Universidad de Las Américas — Quito, Ecuador

- Created a curated **database of 50,000 herbarium records** from tropical forest species of the Americas using web scraping and the Global Biodiversity Information Facility API.
- Developed **machine learning classifiers** to predict the **phenological stages** of forest species from herbarium records, supporting the design of conservation strategies for tropical forests (More details).

CONTACT

sayalaruano.github.io

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SKILLS

Technical

Programming Languages: python, R, BASH, C++, Java

DevOps: GitHub, git, docker

Data Science: jupyter, python, R

Databases and Cloud: MySQL, PostgreSQL, Firebase, neo4j

Languages

Spanish: Native

English: Advanced | C1 | TOEFL iBT 109

The source code is available at sayalaruano/cv.

Last updated on 2026-03-16.

View this CV online at sayalaruano.github.io/cv

2021

Research Assistant

Applied Signal Processing and Machine Learning Research Group - USFQ

Quito, Ecuador

- Contributed to building a **network science and similarity-search method and software** to explore the **chemical space of antimicrobial peptides** and discover new drugs (More details).
- **Advisors**: Yovani Marrero-Ponce, Noel Pérez Pérez

2020 | 2017

Research Intern

Computational and Theoretical Chemistry Group - USFQ

Quito, Ecuador

- Studied **binding specificity** between **repressor proteins and a transcriptional factor** in the jasmonic acid pathway of *Arabidopsis thaliana* using **molecular dynamics** simulations and **machine learning** algorithms (More details).
- **Advisor**: Miguel Angel Méndez Silva

2020 | 2018

Research Intern

Bio-Chemoinformatics Group - Universidad de Las Américas

Quito, Ecuador

- Investigated the impact of **horizontal gene transfer** in the genome of *Streptomyces clavuligerus* using **phylogenetics**, and other bioinformatics tools (More details).
- **Advisors**: Vinicio Armijos, Yunierkis Perez

2019

Research Intern

Tumor Metabolism and Therapeutic Oncology Laboratory - Gwangju Institute of Science and Technology

Gwangju, South Korea

- Performed **density functional theory** and **molecular dynamics** simulations to study the effect of a mutation in the ZN domain of the **CRBN protein** (More details).
- **Advisors**: Miguel Angel Méndez Silva, Steve K. Cho

TEACHING EXPERIENCE

2021

Bioinformatics Boot Camp Co-organizer and Co-instructor

RSG Ecuador and iGEM Ecuador

Virtual

- Designed and taught course materials covering Linux basics, terminal usage, Bash/AWK scripting, and Git/GitHub.

2020

Undergraduate Teaching Assistant

Learning Center - USFQ

Virtual

- Provided online mentorship in biotechnology, mathematics, and engineering subjects to undergraduate students.

2018 | 2016

Undergraduate Teaching Assistant

General Biology Laboratory - USFQ

Quito, Ecuador

- Graded reports, tests, and homework, providing feedback and guidance on course topics.

PUBLICATIONS

Peer reviewed journal articles

- **Ayala-Ruano, S.**, Webel, H., & Santos, A. (2025). **VueGen: Automating the generation of scientific reports**. *Bioinformatics Advances*, vbaf149. doi: doi.org/10.1093/bioadv/vbaf149.
- Ordoñez, J., Tovar, C., Walker, B., Wheeler, J., **Ayala-Ruano, S.**, Aguirre-Carvajal, K., McMahon S., & Cuesta, F. (2025). **Phenological patterns of tropical mountain forest trees across the neotropics: evidence from herbarium specimens**. *Proc. R. Soc. B*, 292 (2041), 20242748. doi: doi.org/10.1098/rspb.2024.2748
- Aguilera-Mendoza, L., **Ayala-Ruano, S.***, Martinez-Rios, F., Chavez, E., García-Jacas, C. R., Brizuela, C. A., & Marrero-Ponce, Y. (2023). **StarPep Toolbox: an open-source software to assist chemical space analysis of bioactive peptides and their functions using complex networks**. *Bioinformatics*, 39 (8), btad506. doi: doi.org/10.1093/bioinformatics/btad506
- **Ayala-Ruano S.**, Marrero-Ponce Y., Aguilera-Mendoza L., Pérez N., Agüero-Chapin G., Antunes A., Aguilar A. (2022). **Network Science and Group Fusion Similarity-Based Searching to Explore the Chemical Space of Antiparasitic Peptides**. *ACS omega*, 7 (50), 46012-46036. doi: doi.org/10.1021/acsomega.2c03398.

- Oña-Chuquimarca, S., **Ayala-Ruano, S.**, Goossens, Pauwels, L., Goossens, A., Leon-Reyes, A., & Méndez, M. A (2020). **The molecular basis of JAZ-MYC coupling, a protein-protein interface essential for plant response to stressors**. *Frontiers in Plant Science*, 11, 1139. doi: 10.3389/fpls.2020.01139. This article was included in the *Frontiers in Plant Science 2020 highlights e-book*. doi: 10.3389/978-2-88966-723-9.
- **Ayala-Ruano, S.**, Santander-Gordón, D., Tejera, E., Perez-Castillo, Y., & Armijos-Jaramillo, V. (2019). **A putative antimicrobial peptide from Hymenoptera in the megaplasmid pSCL4 of Streptomyces clavuligerus ATCC 27064 reveals a singular case of horizontal gene transfer with potential applications**. *Ecology and Evolution*, 9 (5), 2602-2614. doi: 10.1002/ece3.4924.

Editorial journal articles

- Osorio-Mogollon C., Grentzinger V., Olguin-Orellana G.J., **Ayala-Ruano S.**, et al. (2023). **ISCB Student Council Symposium 2021, a virtual global venue: challenges and lessons learned**. *F1000Research*, 12(50). doi: 10.12688/f1000research.129945.1.
- **Ayala-Ruano S.***, Hernandez, F., Ortega, A., Infante, D., Carrascal, D., Sánchez-Luquez, K., & Puche-Quiñonez, R. (2022). **Highlights of the 1st Ecuadorian-Venezuelan Symposium of Young Researchers in Bioinformatics (1SEVJIB)**. *F1000Research*, 11(1086), 1086. doi: 10.12688/f1000research.125381.1.
- Castillo-Vilcahuaman, C., Valdivia C., Osorio-Mogollón C., Silva-Andrade, C., Puche, R., **Ayala-Ruano, S.**, Cuesta-Astroz, Y., Parra, G (2020). **4th ISCB Latin American Student Council Symposium: a virtual and inclusive experience during COVID19 times**. *F1000Research*, 9. doi: 10.12688/f1000research.28330.1.

Preprints

- Vidal E., Phanthanourak, A.L., Gharib, A., Webel, H., Assis, J., **Ayala-Ruano, S.**, Cunha, A.F., & Santos, A. (2025). **ABaCo: Addressing Heterogeneity Challenges in Metagenomic Data Integration with Adversarial Generative Models**. *bioRxiv*. doi: https://doi.org/10.1101/2025.09.22.677692.

*co-first author

SELECTED PRESENTATIONS

- VueGen: Automating the creation of scientific reports. (2025). Oral presentation. **Bioinformatics Open Source Conference (BOSC) - ISMB/ECCB 2025**.
- MicW2Graph: Building a knowledge graph of the wastewater treatment microbiome and its biological context. (2025). Oral presentation. **Conference on Complex Systems 2025**.
- Contributing Guidelines and Codes of Conduct for Open Projects. (2022). Expert talk. **6th cohort of Open Life Science**.
- Exploring the chemical space of antiparasitic peptides and discovery of new leads using network science and similarity searching. (2022). Oral presentation. **International Society for Computational Biology Student Council Webinar series**.
- HerrCompBioinfo: An open-source educational resource of computational tools for Bioinformatics enthusiasts written in Spanish. (2022). Oral presentation. **4th cohort of Open Life Science Graduation**.
- The molecular basis of JAZ-MYC coupling, a protein-protein interface essential for plant response to stressors. (2021). Oral presentation. **6th Brazilian Student Council Symposium: Omics and Data Science**.
- In silico detection of horizontal gene transfer in Streptomyces clavuligerus. (2020). Oral presentation. **International Society for Computational Biology Student Council Webinar series**.
- Structural changes due to a mutation in Cereblon might be a cause for intellectual disability. (2019). Poster presentation. **Global Intern Program - Gwangju Institute of Science and Technology**.

HONORS AND AWARDS

2024

MaCSBio Thesis Award

Maastricht University — Maatricht, the Netherlands

- Recognizes one MSc System Biology thesis based on **scientific merit, thesis quality, and academic performance**.

2024

Visit Grant

Danish Data Science Academy — Copenhagen, Denmark

- Covered **travel and accommodation expenses** for MSc thesis internship.

2024 | 2022

Cum Laude Graduation Distinction

Maastricht University — Maatricht, the Netherlands

- Awarded to students with a **GPA ≥ 8/10**.

2022

UM Holland-High Potential Scholarship

Maastricht University — Maatricht, the Netherlands

- Full scholarship covering **tuition and stipend** support for highly talented non-EU students admitted to a master's program.

2021

Best oral presentation award

6th Brazilian Student Council Symposium: Omics and Data Science — Virtual

2021

"For More Data on Labor Informality" Innovation Challenge Award

Datalat, PNUD Ecuador, and the International Labour Organization — Virtual

- Selected among 39 proposals from 80 interdisciplinary teams to design a technological solution for **collecting labor informality data** in Ecuador (More details).

2020 | 2016

Chancellor's Honor List and Magna Cum Laude Graduation Distinction

Universidad San Francisco de Quito — Quito, Ecuador

- Recognized students with a **GPA ≥ 3.7/4**.

2019

2nd RSG-Colombia Symposium travel award

RSG Colombia — Ibagué, Colombia

- Covered **travel expenses** to attend the event.

2019

Global Intern Program Scholarship

Gwangju Institute of Science and Technology — Gwangju, South Korea

- Covered **accommodation and stipend** for an 8-week research internship with training and mentoring.

LEADERSHIP AND SERVICE

Current | 2022

Streamlit Creators Program

Streamlit — Virtual

- Developed multiple web applications using the **Streamlit Python package** and became part of the creators community.

Current | 2021

Open Life Science (OLS) program

Open Life Science — Virtual

- Led the HerrCompBioinfo project during the 4th cohort, creating an **open-source educational resource** of computational tools for **bioinformatics enthusiasts** written in Spanish.
- Mentored a team in the 5th cohort to create a computer vision tool for improving cancer diagnosis in Cameroon.
- Delivered a talk about contributing guidelines and codes of conduct for open projects in the 6th cohort.

2023 | 2020

Regional Student Group (RSG) Ecuador

International Society for Computational Biology Student Council — Virtual

- Co-founder, built a **community of bioinformatics students** and researchers in Ecuador.
- Co-chair of the 1st Ecuadorian-Venezuelan Symposium of Young Bioinformatics Researchers.
- Fellowship committee chair, 17th Student Council Symposium.
- Contributed to the program and fellowships committees of the 4th ISCB Latin American Student Council Symposium.

2021

Saturdays.AI Quito 2021

Saturdays.AI Quito — Virtual

- Led a project developing a maize disease detector based on convolutional neural networks.

Note: I have developed additional personal projects about **bioinformatics, network science, machine/deep learning, software and web development**, and other topics.